Fact Families and Inverse Operations

Australian Curriculum v9.0 · Year 6 Maths
(Year 5 review skill)
AC9M5A01

🎒 Back to School 2027 Warm-Up

Name: _____ Date: _____

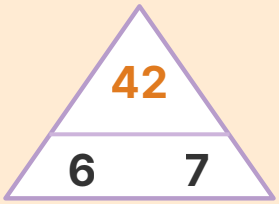
LEARNING INTENTION

I can use multiplication and division as inverse operations to build fact families and to check my answers.

I CAN…

- ✓ write a fact family from three numbers
- ✓ find a missing number using the inverse
- ✓ check an answer using the inverse operation

KEY STRATEGY AND WORKED EXAMPLE



Inverse operations undo each other. Multiplication and division are a pair.
6 × 7 = 42 7 × 6 = 42
42 ÷ 6 = 7 42 ÷ 7 = 6
Checking: 156 ÷ 12 = 13 → check 13 × 12 = 156 ✓

In a division fact, the **biggest number** (the product) always comes first: **42 ÷ 6 = 7, never 6 ÷ 42.**

A. TRY TOGETHER – USE THE STRATEGY

1. Complete the family for 8, 9 and 72:
   8 × 9 = _____ 9 × 8 = _____
   72 ÷ 8 = _____ 72 ÷ 9 = _____
2. 7 × _____ = 56
3. _____ ÷ 6 = 9
4. 63 ÷ _____ = 7
5. Write both division facts for 25 × 4 = 100:
   _____ _____

B. CORE PRACTICE – CALCULATE AND CHECK

6. _____ × 8 = 96
7. 144 ÷ _____ = 12
8. _____ ÷ 15 = 6
9. 13 × 7 = 91, so 91 ÷ 13 = _____
10. **Check with the inverse.** Gus says 252 ÷ 12 = 21. Use multiplication to check. Is he correct?

    _____

    _____
11. **Check with the inverse.** Ana says 18 × 15 = 280. Use division to check. Is she correct? If not, what is the right answer?

    _____

    _____
12. 54 × _____ = 540
13. _____ × 25 = 600

C. CHALLENGE – REASON AND JUSTIFY

14 **216 students** are sorted into **8 equal teams**. Write a number sentence to find how many students are in each team, then use the inverse to check your answer.

15 **Error Analysis.** Noah wrote this fact family: 9 × 6 = 54, 6 × 9 = 54, 54 ÷ 9 = 6, 9 ÷ 54 = 6. Which fact is wrong? Explain why, then write the correct fact.

16 Zara says, “Every fact family has four facts.” Use 7 × 7 = 49 to decide if she is right. Explain your thinking.

Watch Out! Only multiplication is the same either way round (6 × 7 = 7 × 6). Division is not: 42 ÷ 6 is not the same as 6 ÷ 42.

💡 **Parent Tip:** When your child gets an answer, ask: “How could you check that using the opposite operation?” Make it a habit for homework, shopping and cooking sums.

🎯 My Term 1 Goal: This term I want to get better at _____

Answer and Teacher Page

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ANSWERS

1. 8 × 9 = 72, 9 × 8 = 72, 72 ÷ 8 = 9, 72 ÷ 9 = 8

2. 8

3. 54

4. 9

5. 100 ÷ 25 = 4 and 100 ÷ 4 = 25

6. 12

7. 12

8. 90

9. 7

10. Yes. 21 × 12 = 252, so 252 ÷ 12 = 21 is correct.

11. No. 280 ÷ 15 is not 18 (15 × 18 = 270). The correct answer is 270.

12. 10

13. 24

14. 216 ÷ 8 = 27 students per team. Check: 27 × 8 = 216 ✓

15. 9 ÷ 54 = 6 is wrong. In division the biggest number comes first. Correct: 54 ÷ 6 = 9.

16. Zara is not right. 7 × 7 = 49 has only two facts: 7 × 7 = 49 and 49 ÷ 7 = 7, because both factors are the same number.

TEACHER NOTES

Common misconception: writing the division fact the wrong way round (small number first) or assuming division is commutative.
Support: use the fact triangle on page 1. Cover one number at a time to find the missing one.
Challenge marking: Q14 correct number sentence plus an inverse check. Q15 must mention that the product / biggest number comes first. Q16 look for the idea that a repeated (square) factor gives only two facts.
Australian Curriculum v9.0: AC9M5A01 (Year 5 review): recognise and explain the connection between multiplication and division as inverse operations and use this to develop families of number facts.